

E-COMMERCE SALES & PROFITABILITY ANALYSIS

Business Performance Report

End-to-end analytics using Python, SQL Server and Power BI

Total Sales	Total Profit	Orders	Customers
\$2,297,201	\$286,397	5,009	793

Overall profit margin: 12.5% | Dataset: 9,994 transaction rows

1. Executive Summary

This project evaluates the sales and profitability performance of a retail/e-commerce business case using the Superstore dataset. The analysis was structured around a practical management question: **where is the business generating revenue, where is it generating profit, and where are margins being eroded?**

The workflow combined Python for data preparation and quality checks, SQL Server for structured business analysis, and Power BI for interactive management reporting.

Key findings

- The business generated **\$2,297,201** in sales and **\$286,397** in profit, with an overall margin of **12.5%**.
- Sales increased from \$484,247 in 2014 to \$733,215 in 2017; the largest annual sales increase occurred in 2016 (29.5%).
- Technology was the largest sales category, while Furniture produced a much weaker margin relative to its sales base.
- Tables, Bookcases and Supplies were loss-making sub-categories and should be reviewed for pricing, discounting and product economics.
- West was the strongest region by both sales and profit, while Central had the weakest regional profit margin.
- Higher discounts were associated with weaker profitability; discounts of 30% and above produced negative aggregate margins in this dataset.

Management implication: the next opportunity is profitable growth—controlling margin leakage while continuing to build sales.

2. Business Objective & Analytical Approach

Business problem

Management requires visibility into sales growth, profitability, product/category performance, customer segments, regional performance and the effect of discounts and shipping choices.

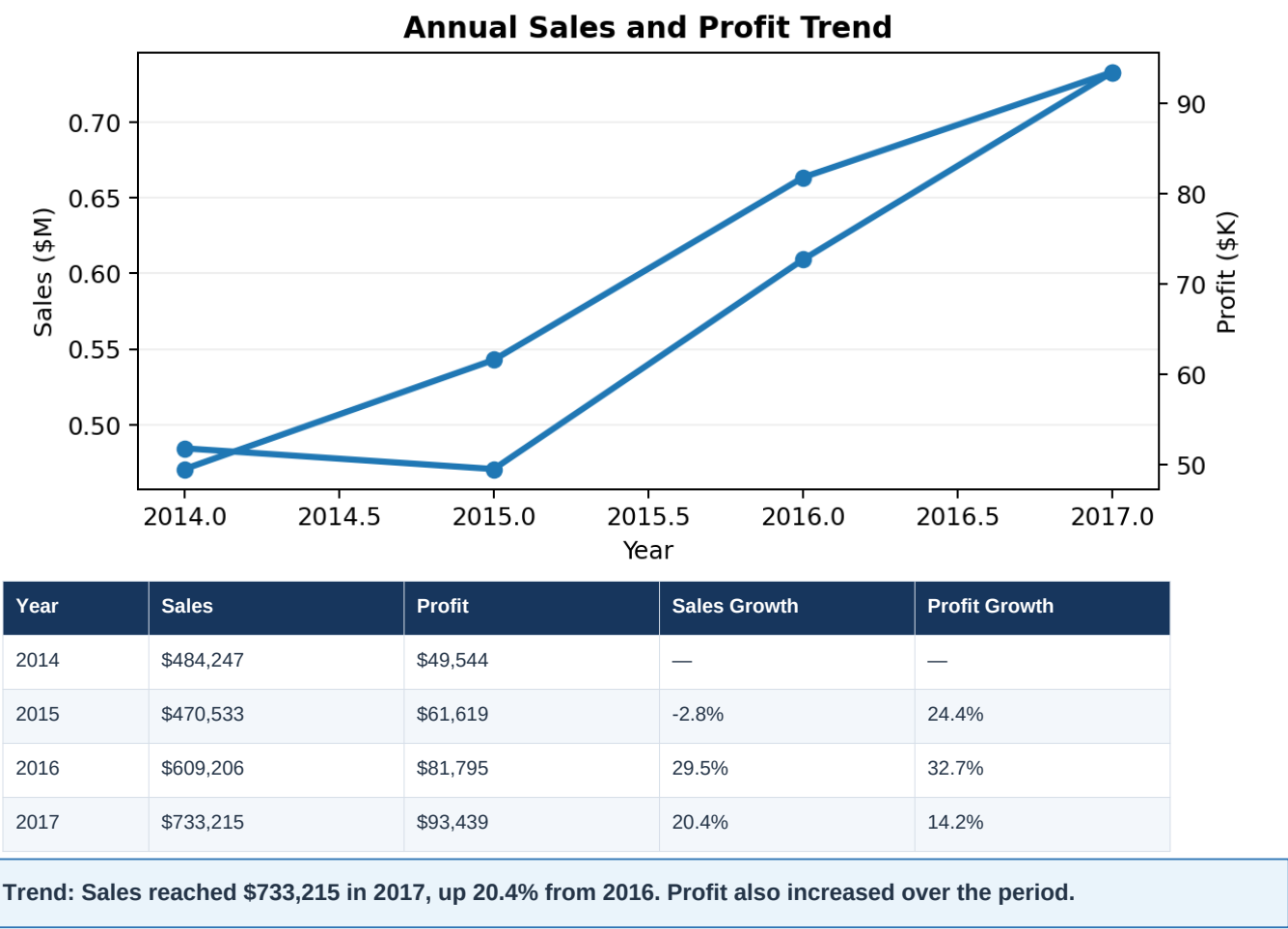
Analytical workflow

Stage	Tool	Purpose
Data preparation	Python / Pandas	Load data, inspect quality, convert dates and create year/month fields.
Business analysis	SQL Server	Calculate sales, profit, margins, trends, products, categories, segments, regions, discounts and shipping
Visualization	Power BI	Build an interactive management dashboard with KPI and performance views.
Decision support	Business report	Translate results into findings, implications and recommendations.

Data quality

Characteristic	Result
Original records	9,994
Original columns	21
Prepared columns	24
Missing values	No missing values identified in the final prepared dataset
Date preparation	Order Date converted to datetime; Year, Month and Month Name derived
SQL table	ecommerce_sales

3. Sales & Profitability Performance

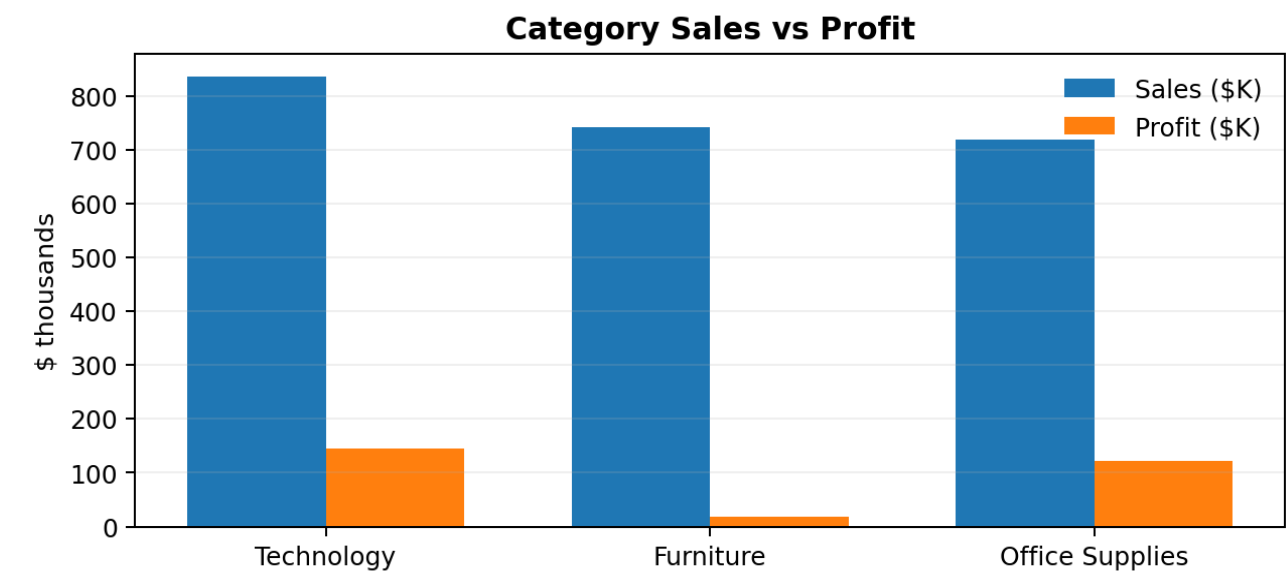


Monthly highlights

Highest sales: **November 2017** (\$118,448). Lowest sales: **February 2014** (\$4,520). Highest monthly profit: **December 2016** (\$17,885). Lowest monthly profit: **January 2015** (\$-3,281).

The monthly pattern shows meaningful seasonality and volatility, supporting stronger planning around peak periods.

4. Product & Category Performance



Category	Sales	Profit	Margin
Technology	\$836,154	\$145,455	17.4%
Furniture	\$742,000	\$18,451	2.5%
Office Supplies	\$719,047	\$122,491	17.0%

Technology led sales at \$836,154 and generated \$145,455 profit. Furniture generated \$742,000 sales but only \$18,451 profit, a 2.5% margin.

Sub-category profitability

Sub-Category	Sales	Profit	Margin
Copiers	\$149,528	\$55,618	37.2%
Phones	\$330,007	\$44,516	13.5%
Accessories	\$167,380	\$41,937	25.1%
Paper	\$78,479	\$34,054	43.4%
Binders	\$203,413	\$30,222	14.9%
Chairs	\$328,449	\$26,590	8.1%
Storage	\$223,844	\$21,279	9.5%
Appliances	\$107,532	\$18,138	16.9%

Loss-making sub-categories

Sub-Category	Sales	Profit	Margin
Tables	\$206,966	\$-17,725	-8.6%
Bookcases	\$114,880	\$-3,473	-3.0%
Supplies	\$46,674	\$-1,189	-2.5%

Action point: investigate loss-making groups at SKU level for pricing, discounting, cost-to-serve and assortment decisions.

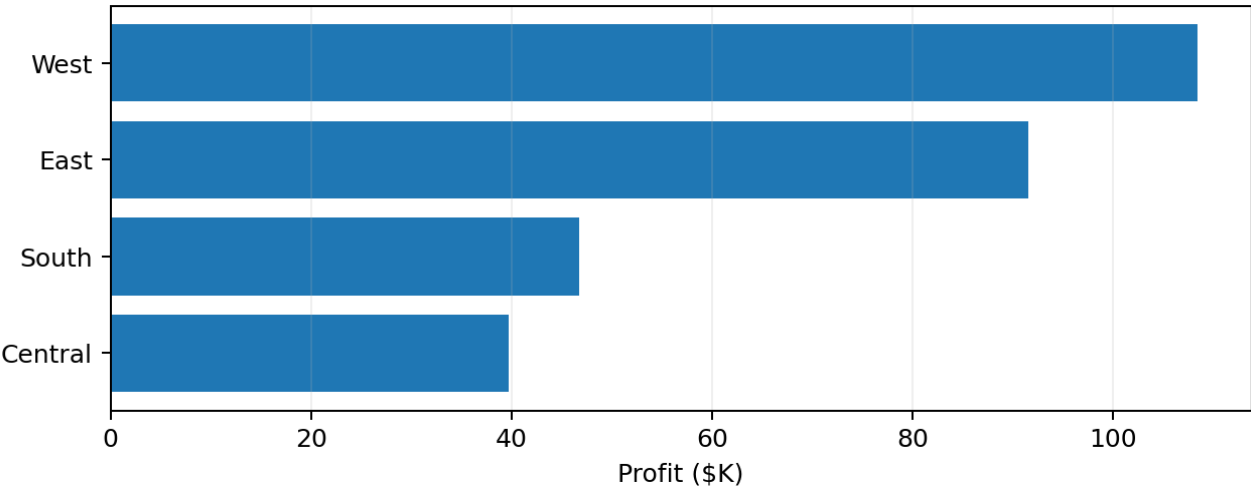
5. Customer Segment & Regional Performance

Customer segments

Segment	Sales	Profit	Customers	Margin
Consumer	\$1,161,401	\$134,119	409	11.5%
Corporate	\$706,146	\$91,979	236	13.0%
Home Office	\$429,653	\$60,299	148	14.0%

Consumer contributed the largest absolute sales and profit because of its scale, while Home Office recorded the highest segment margin.

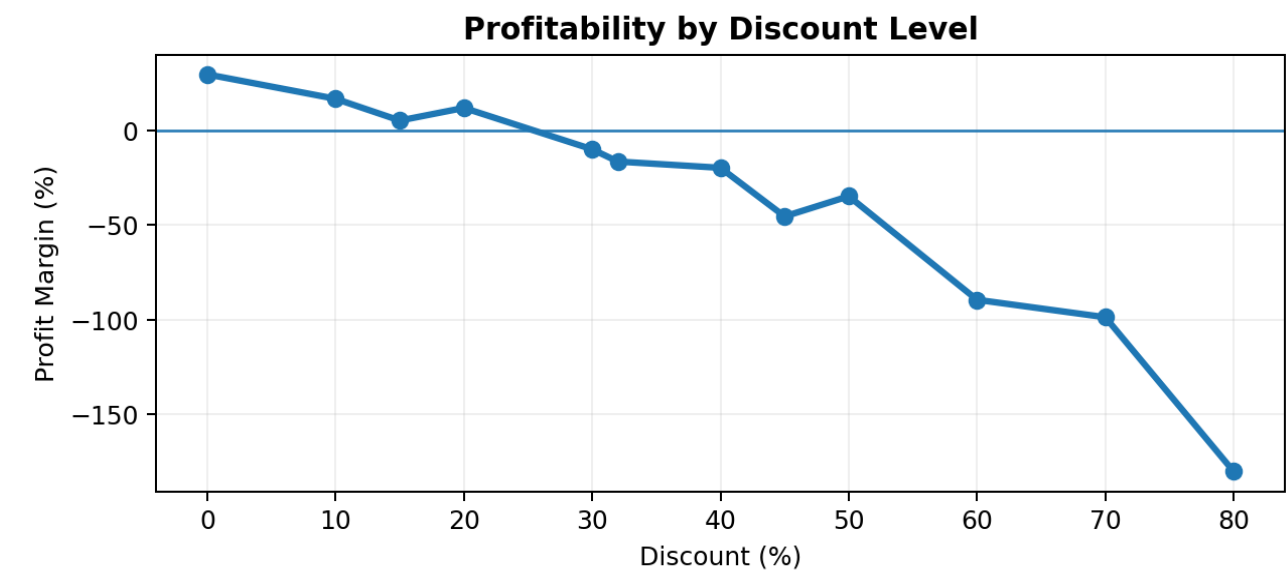
Profit by Region



Region	Sales	Profit	Margin
West	\$725,458	\$108,418	14.9%
East	\$678,781	\$91,523	13.5%
Central	\$501,240	\$39,706	7.9%
South	\$391,722	\$46,749	11.9%

Regional insight: West is the strongest overall market. Central's lower margin suggests an opportunity to examine its product mix, pricing and discount behavior.

6. Commercial Drivers: Discounts, Shipping & Products



Profitability deteriorates materially as discount levels rise. Aggregate margin becomes negative at 30% discount and remains negative at higher discount levels in the dataset.

Discount	Sales	Profit	Margin
0%	\$1,087,908	\$320,988	29.5%
10%	\$54,369	\$9,029	16.6%
15%	\$27,559	\$1,419	5.1%
20%	\$764,594	\$90,337	11.8%
30%	\$103,227	\$-10,369	-10.0%
32%	\$14,493	\$-2,391	-16.5%
40%	\$116,418	\$-23,057	-19.8%
45%	\$5,485	\$-2,493	-45.5%
50%	\$58,919	\$-20,506	-34.8%
60%	\$6,645	\$-5,945	-89.5%
70%	\$40,620	\$-40,075	-98.7%
80%	\$16,964	\$-30,539	-180.0%

Shipping performance

Ship Mode	Orders	Sales	Profit	Margin
Standard Class	2,994	\$1,358,216	\$164,089	12.1%
Second Class	964	\$459,194	\$57,447	12.5%
First Class	787	\$351,428	\$48,970	13.9%
Same Day	264	\$128,363	\$15,892	12.4%

Standard Class is the dominant shipping mode by orders and sales. First Class recorded the highest margin among the shipping methods.

Top products by sales

Rank	Product	Sales	Profit
1	Canon imageCLASS 2200 Advanced Copier	\$61,600	\$25,200
2	Fellowes PB500 Electric Punch Plastic Comb Bindin...	\$27,453	\$7,753
3	Cisco TelePresence System EX90 Videoconferencing ...	\$22,638	\$-1,811
4	HON 5400 Series Task Chairs for Big and Tall	\$21,871	\$0
5	GBC DocuBind TL300 Electric Binding System	\$19,823	\$2,234
6	GBC Ibimaster 500 Manual ProClick Binding System	\$19,024	\$761
7	Hewlett Packard LaserJet 3310 Copier	\$18,840	\$6,984
8	HP Designjet T520 Inkjet Large Format Printer - 2...	\$18,375	\$4,095
9	GBC DocuBind P400 Electric Binding System	\$17,965	\$-1,878
10	High Speed Automatic Electric Letter Opener	\$17,030	\$-262

7. Management Recommendations

1. Protect margin in high-discount transactions

Introduce discount guardrails by category and product. High discounts should require evidence that incremental volume compensates for margin loss.

2. Fix loss-making sub-categories

Prioritize Tables, Bookcases and Supplies for SKU-level review of pricing, supplier cost, discounting and assortment.

3. Improve Furniture economics

Furniture has substantial sales but a weak aggregate margin. Identify the products and discount levels driving margin leakage.

4. Replicate successful regional practices

Use West as a benchmark and compare its product mix, customer composition and discount behavior with lower-margin regions.

5. Focus on profitable customer growth

Use segment-specific pricing and promotions rather than relying only on sales volume; Home Office demonstrates that smaller segments can still have stronger margins.

6. Plan around seasonal demand

Use the stronger year-end demand pattern to plan inventory, promotions and capacity ahead of peak periods.

Priority: shift the management focus from sales growth alone to profitable sales growth.

8. Conclusion

The analysis provides a complete view of revenue, profit, products, customers, regions, discounts and shipping. The business shows a strong growth trajectory, but revenue growth does not automatically translate into healthy profitability.

The strongest opportunity is therefore to improve **margin quality**: control excessive discounting, address loss-making product groups, and use stronger regions and customer segments as benchmarks for profitable growth.

Project deliverables

Component	Output
Python	Data loading, quality checks, date preparation and SQL Server data loading
SQL Server	12 business questions covering trends, products, categories, segments, regions, discounts and shipping
Power BI	Interactive management dashboard with KPI, trend, category, region and customer views
Business Report	Management-focused findings, implications and recommendations

Note: This report is based on the supplied Superstore dataset and completed project analysis. Figures represent the dataset and are not claims about a named real-world company.